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A flexible Weibull geometric distribution with characterizations and its parameter estimation

Ahmadreza Zanboori[1], Ehsan Zanboori[1]✉, Hamid Parvin[2] & Mohammadreza Mahmoudi[3,4]✉

The current paper introduces a new version of the Weibull-geometric distribution, which offers a more flexible model for lifetime data. The statistical properties of the proposed distribution—such as the probability density function, reliability and failure rate functions, quantiles, and moments—are discussed. The EM algorithm is used to compute the asymptotic variances and covariances of the parameters. Point estimators of the unknown parameters, under various symmetric and asymmetric loss functions, are obtained using the Bayesian framework and the Markov Chain Monte Carlo (MCMC) technique. Furthermore, using the importance sampling procedure, the highest posterior density (HPD) credible intervals for the parameters are derived. Maximum likelihood and Bayesian estimators are also employed to compute the shrinkage preliminary test estimators. Consequently, a simulation study is conducted to evaluate the performance of all proposed estimation methods. Finally, a real data set is analyzed to demonstrate the effectiveness and flexibility of the new distribution.

Keywords Bayesian estimation, EM algorithm, Failure rate function, Shrinkage preliminary test estimators, Asymptotic normality, New Weibull-geometric distribution

Lifetime data modeling and analysis is an important aspect of statistical tasks in many applied sciences, including medicine, engineering, and finance. In the literature, several distributions have been proposed to model such types of data by combining some beneficial life distributions. In the literature, a few appropriations have been proposed to show such kinds of information by consolidating some valuable life distributions. Adamidis & Loukas[1] suggested a two-parameter exponential-geometric (EG) distribution, which was generated by combining an exponential and a geometric distribution. Bidram & Behboodian[2] presented a novel generalized exponential geometric distribution, which is an extension of the EG model with DFR and increasing failure rate (IFR) functions. Furthermore, Adamidis et al.[3] presented the extended exponential-geometric (EEG) distribution, which generalizes the EG distribution and addresses some of its statistical properties as well as its reliability features. The hazard function of the EEG distribution can be monotone decreasing, increasing, or constant. Rasekhi et al.[4] call the modified exponential (ME) distribution a generalization of the exponential distribution by combining the extended exponential distribution and the generalized exponential distribution. Zanboori et al.[5] proposed a three-parameter lifetime distribution with increasing and decreasing hazard functions. All of the above-mentioned compounding distributions are based on the Exponential distribution and have a monotone failure rate.

Statistical distributions play a vital role in modeling and analyzing lifetime data in various applied fields such as medicine, engineering, and reliability. Many researchers have proposed new families of distributions or extended existing ones to better fit empirical data and provide more flexibility in capturing different hazard rate behaviors. Among them, mixed and compounded distributions have shown promising results in improving the modeling accuracy of lifetime phenomena.

Recently, several efforts have been made to develop and study new statistical distributions and estimation methods applicable to real-world problems. For instance, optimal estimation under censoring schemes[6], high-dimensional Bayesian shrinkage techniques[7], applications in tumor dynamics through optimal control systems[8], and modifications to classical distributions for greater flexibility[5] have been investigated and demonstrated significant practical benefits.

In this study, we introduce a new flexible distribution, called the NWG (New Weibull-Geometric) distribution, and explore its statistical properties, estimation methods, and applications. We assess its performance by

[1]Department of Mathematics, NoM. C., Islamic Azad University, Noorabad Mamasani, Iran. [2]Department of Computer Engineering, NoM. C., Islamic Azad University, Noorabad Mamasani, Iran. [3]Department of Statistics, Faculty of Science, Fasa University, Fasa, Iran. [4]Research Institute of Statistics, Data Mining and Data Science, Fasa University, Fasa, Iran. ✉email: ehsan.zanboori@iau.ac.ir; Mahmoudi.m.r@fasau.ac.ir

analyzing both simulated and real-world data, with a focus on remission times of bladder cancer patients. The results show that the NWG model offers a better fit compared to classical alternatives and provides reliable parameter estimation through various techniques, including MLE, Bayesian methods, and shrinkage-type procedures.

The Weibull distribution is one of the most widely used distributions for modeling lifetime data. It has gotten wide applications in expansive zones, including reliability engineering, survival analysis, hydrology, meteorology, and insurance.

Numerous distributions have been proposed inside the literature to extend the Weibull distribution. Wanbo and Daimin[9] proposed the Weibull-Poisson (WP) distribution, which combines the Weibull and Poisson distributions. The hazard function of the WP distribution can be increasing, decreasing, upside-down, bathtub-shaped, or unimodal. Furthermore, Barreto-Souza et al.[10] proposed a Weibull-geometric (WG) distribution, which combines the Weibull and geometric distributions. Several properties of the WG distribution were also investigated. It can additionally be used in a range of problems in distinctive areas such as public health, biomedical studies, and industrial reliability and survival analysis. Motivated to develop a basic and adaptable model to explain phenomena with monotone hazard rates, which are popular in reliability and biological research. The aim of this paper is to introduce a new paradigm as a rival to some existing distributions for fitting on some data sets. In this regard, we propose a new Weibull-geometric (NWG) distribution, similar to that proposed by Barreto-Souza et al.[10], that outperforms some well-known distributions, particularly that proposed by Barreto-Souza et al.[10], in terms of some statistical criteria when tested on some real data sets.

The aim of this paper is to accomplish two parts. We proposed a novel compound distribution of the Weibull and geometric distributions in the first part. The statistical properties of the proposed distribution are discussed, including the probability density function, reliability and failure rate functions, quantiles, and moments. In the second section, we look at the problem of parameter estimation for various parameter values using both classical and Bayesian paradigms. Under the respective approach, we also compute the 95% asymptotic confidence interval and the highest posterior density interval estimators. Additionally, maximum likelihood and Bayesian shrinkage estimators are developed.

The organization of the present paper is as follows. Section"The new Weibull-Geometric distribution"proposes and illustrates a novel distribution with its probability density function (pdf). Many structural properties of the new distribution were examined in Sect."Structural properties". In addition, the cumulative distribution function (cdf), survival and hazard functions, quantiles, and moments were studied. This section also covers the expressions and computations for density of order statistics and their moments. The Fisher knowledge matrix and maximum likelihood estimate of the parameters are discussed in Sect."Maximum likelihood estimation". In Sect."Bayesian estimation", we looked at estimation strategies in a Bayesian system. The shrinkage preliminary test estimators are discussed in Sect."Shrinkage preliminary test estimator". Sections"Shrinkage preliminary test estimator"and"Simulation study"discuss the real data set and the simulation analysis, respectively. Finally, in Sect."Conclusions", some concluding remarks are presented.

The new Weibull-Geometric distribution

It is supposed that $W_1, W_2, \ldots, W_z$ are independent, random, and identically distributed (iid) variables following the Weibull distribution $W(a,b)$ with pdf:

$$g\left(w;a,b\right) = ab^a w^{a-1} e^{-(bw)^a}, \quad w>0, \quad a,b>0 \tag{1}$$

where $a>0$ and $b>0$ indicate the shape and scale parameters, respectively and Z has a geometric distribution. The probability mass function of $P\left(z;p\right) = (1-p)p^{z-1}$ with the assumption that Z and W are independent. By assuming $X = \max(W_1, W_2, \ldots, W_z)$; Then:

$$f\left(x|z;a,b\right) = ab^a z x^{a-1} e^{-(bx)^a} \left(1-e^{-(bx)^a}\right)^{z-1}, \quad x>0 \tag{2}$$

Furthermore, the marginal pdf of X is as follows:

$$f\left(x;a,b,p\right) = \frac{ab^a(1-p)x^{a-1}e^{-(bx)^a}}{\left[1-p\left(1-e^{-(bx)^a}\right)\right]^2}, \quad x>0 \tag{3}$$

where x > 0 and a,b > 0, p ϵ(0,1). The random variable, X with the density function expressed in Eq. (3) supposedly has a new Weibull-geometric (*NWG*) distribution denoted by $X \sim NWG(a,\mathbf{b},p)$. In case of $a = 1$, the *NWG* distribution is reduced to an *EG* distribution

Proposition 1 *Considering the NWG distribution with the pdf of* Eq. (3), *the properties below are achieved*:

(a) The NWG *distribution is reduced to a Weibull distribution with two parameters as* $p \to 0$.

(b) The density function of NWG *distribution is monotone decreasing when* $0 < a \leq 1$ *and is unimodal when* $a>1$. *Here, the mode is equal to* $u_0^{\frac{1}{a}}/b$, *in which* u_0 *is a root of this nonlinear Equation*: $\frac{1-p(1+e^{-u_0})}{1-p(1-e^{-u_0})} = (a-1)/a$.

Proof (a) If p approaches 0, then

$\lim_{p\to 0} f(x;a,b,p) = \lim_{p\to 0} \frac{ab^a(1-p)x^{a-1}e^{-(bx)^a}}{\left[1-p\left(1-e^{-(bx)^a}\right)\right]^2} = ab^a x^{a-1} e^{-(bx)^a}.$
Afterward, the NWG distribution is reduced to a two-parameter Weibull distribution.
(b) The first derivation of the density function is obtained via

$$f'(x) = \frac{ab(1-p)x^{a-2}e^{-(bx)^a}}{\left[1-p\left(1-e^{-(bx)^a}\right)\right]^2} \left\{(a-1) - a(bx)^a \left[1 - \frac{2pe^{-(bx)^a}}{\left(1-p\left(1-e^{-(bx)^a}\right)\right)}\right]\right\}.$$

It is supposed that $\left\{a-1-a(bx)^a\left[1-\frac{2pe^{-(bx)^a}}{\left(1-p\left(1-e^{-(bx)^a}\right)\right)}\right]\right\} = a-1-ah(u)$, where $u=(bx)^a$ and $h(u) = u\left[\frac{1-p(1+e^{-u})}{1-p(1-e^{-u})}\right]$. The sign of $f'(x)$ is the sign of the curly brackets. Subsequently, when the curly brackets are zero, $f'(x)$ is zero. □

The NWG density function plots for selected parameter values are presented in Fig. 1. It demonstrates the function's flexibility in modeling lifetime data. The NWG density tends toward zero for all parameter values, as $x \to \infty$.

Structural properties

This section discusses some basic structural properties of the NWG distribution. The mentioned properties include the distribution function, the corresponding hazard rate function, the shapes of this hazard rate function, order statistics, the rth moment, moment generating function, Renyi entropy, and the Shannon entropy.

Distribution and hazard functions

The cdf of X is given by

$$F(x) = \frac{(1-p)\left(1-e^{-(bx)^a}\right)}{1-p\left(1-e^{-(bx)^a}\right)} \tag{4}$$

and the survival function by

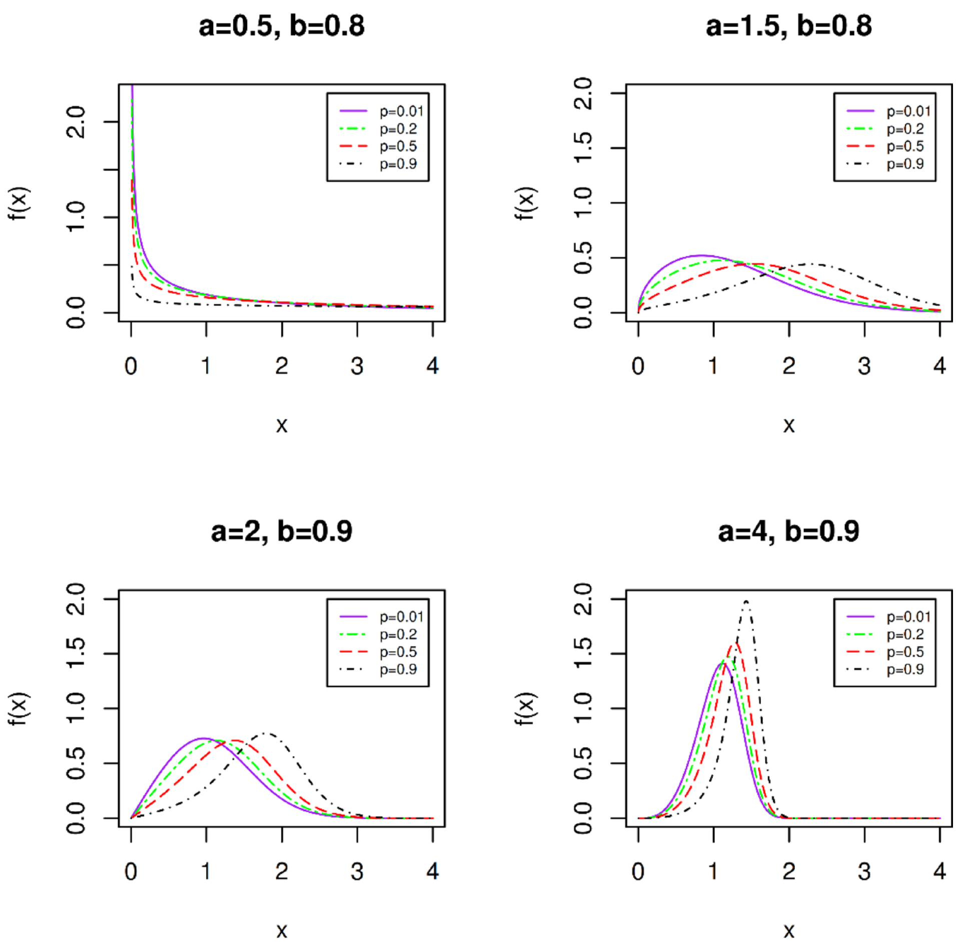

Fig. 1. Probability density function of NWG distribution for selected parameters values.

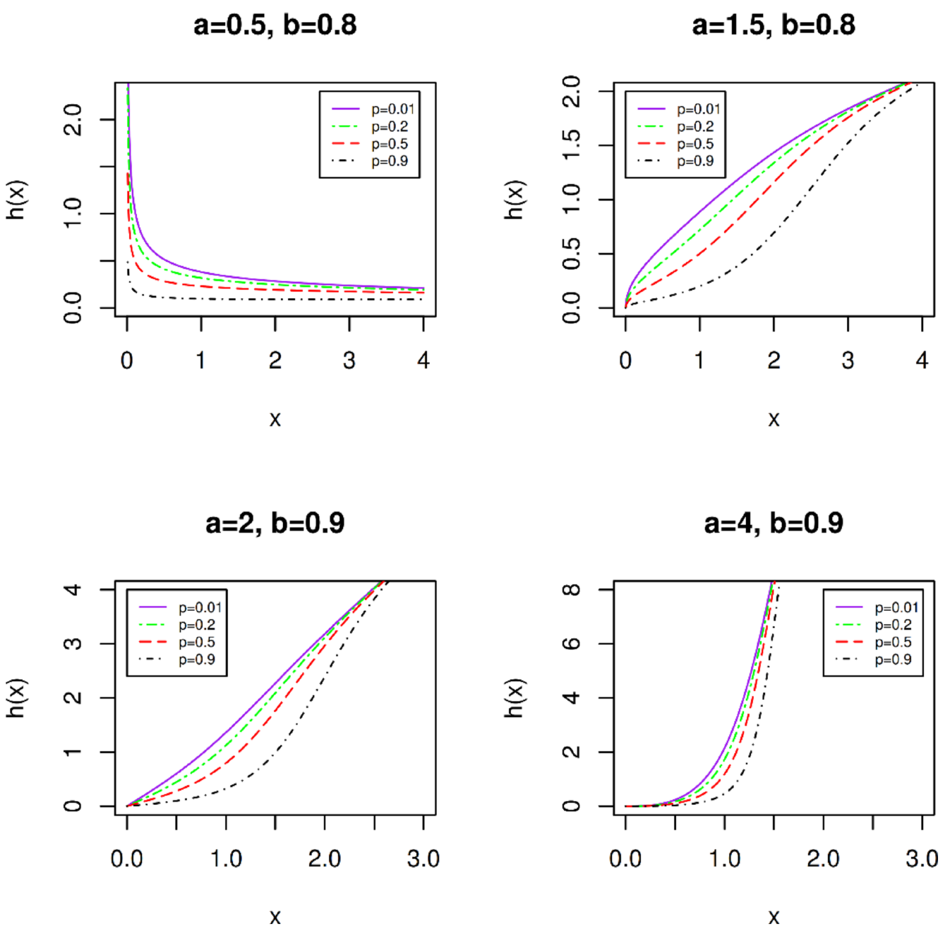

Fig. 2. Hazard rate function of the NWG distribution for selected parameters values.

$$S\left(x\right)=\frac{e^{-(bx)^{a}}}{1-p(1-e^{-(bx)^{a}})}\tag{5}$$

where, $x>0,\ a,b>0$ and $p\in\ (0{,}1)$. The hazard function is a significant characteristic of lifetime modeling, and can be expressed as the following equation for a continuous distribution with cdf of $F\ \left(x\right)$ and pdf of $f\left(x\right)$:

$$h\left(x\right)=\lim_{\varDelta x\to 0}\frac{P(X<x+\varDelta x|X>x)}{\varDelta x}=\frac{f\left(x\right)}{S\left(x\right)}.\tag{6}$$

It can be expressed that the decreasing or increasing property of $h(x)$ depends on the parameters that it is based upon them

From Eqs. (4) and (5), it can be conveniently verified that the hazard rate function is presented by

$$h\left(x\right)=\frac{ab^{a}(1-p)x^{a-1}}{1-p(1-e^{-(bx)^{a}})},\quad x>0.\tag{7}$$

The following function is defined based on Glaser's theorem:

$$\eta\ \left(x\right)=-\frac{f'\ \left(x\right)}{f\left(x\right)}\tag{8}$$

It can be readily demonstrated that:

$$\eta\ \left(x\right)=ab^{a}x^{a-1}\left(1-2pe^{-(bx)^{a}}\left[1-p(1-e^{-(bx)^{a}})\right]^{-1}\right)-\frac{a-1}{x},$$

And its first derivative is

$$\eta'(x) = \frac{1}{x^2}\left[(a-1) + a\,(bx)^a\left\{(a-1)\left(\frac{1-p(1+e^{-(bx)^a})}{1-p(1-e^{-(bx)^a})}\right) - \frac{2p(1-p)a\,(bx)^a e^{-(bx)^a}}{(1-p\,(1-e^{-(bx)^a}))^2}\right\}\right].$$

If $0 < a \leq 1$, then $\eta'(x) < 0$ for all x values. Subsequently, the failure rate function will be decreasing based on the Glaser's theorem. If $a > 1$ and $p \leq \frac{a}{a-1}$, then $\eta'(x) = \frac{1}{x^2}\left[aug(u) + (a-1)\right]$, where $u = (bx)^a$ a and $g(u) = \left[(a-1)\ \frac{1-p\left(1+e^{-u}\right)}{1-p\left(1-e^{-u}\right)} + 2ap\,(1-p)\ \frac{ue^{-u}}{\left(1-p\left(1-e^{-u}\right)\right)^2}\right]$. Using the simple inequality of $\frac{au}{a-1} \leq u/p < e^u/p$ as $p \leq a/(a-1)$, the inequality of $g(u) > 0$ is obtained, then the inequality of $\eta'(x) > 0$ governs for all x values. Consequently, it is concluded that, based on the Glaser's theorem, $h(x)$ is in increasing state. Figure 2 shows the $h(x)$ plot for different parameters. As is shown, the failure rate function of NWG distribution has flexibility.

Quantiles and moments

The γ quantile (x_γ) of the NWG is achieved by inverting Eq. (4), thereby

$$x_\gamma = F^{-1}(\gamma) = -b^{-1}\left[\log\left(\frac{(1-p)\,(1-\gamma)}{1-p(1-\gamma)}\right)\right]^{1/a}. \tag{9}$$

In particular, the median of X is as follows:

$$x_{0.5} = -b^{-1}\left[\log\left(\frac{1-p}{2-p}\right)\right]^{1/a}.$$

An overall expression for the rth moment of the NWG distribution can be obtained through analysis considering the Lerch's transcendent function denoted by $\Phi(\delta, s, c)$, which it is expressed as:

$$\Phi(\delta, s, c) = \{\Gamma(s)\}^{-1}\int_0^\infty t^{s-1}e^{-ct}(1-\delta\, e^{-t})^{-1}dt, \quad \delta < 1,\ c, s > 0. \tag{10}$$

Proposition 2 *It is supposed that* X *stands for a random variable from an* NWG *distribution with parameters* $a, b > 0$ *and* $p \in (0,1)$ *with pdf illustrated by* Eq. (3), *and* $r \in \mathbb{N}$, *subsequently*

$$E(X^r) = \frac{rb^{-r}}{a\,(1-p)}\Gamma\left(\frac{r}{a}\right)\Phi\left(\frac{p}{p-1}, \frac{r}{a}, 1\right), \tag{12}$$

where $\Phi(\delta, \mathrm{s}, \mathrm{c})$ *is the Lerch's transcendent function.*

Proof By definition, $E(X^r) = \int_0^\infty x^r f(x)\,dx$. Using Fubinni theorem and convenient integration, E(X^r) can be represented as follows:

$$E(X^r) = \int_0^\infty \left(\int_0^x rt^{r-1}dt\right) f(x)\,dx = r\int_0^\infty t^{r-1}\left(\int_t^\infty f(x)\,dx\right)dt$$
$$= r\int_0^\infty t^{r-1}S(t)\,dt = r\int_0^\infty \frac{t^{r-1}e^{-(bt)^a}}{1-p\left(1-e^{-(bt)^a}\right)}dt$$
$$= \frac{rb^{-r}}{a(1-p)}\int_0^\infty \frac{u^{r/a-1}e^{-u}}{1-\left(\frac{p}{p-1}\right)e^{-u}}du = \frac{rb^{-r}}{a(1-p)}\Gamma\left(\frac{r}{a}\right)\Phi\left(\frac{p}{p-1}, \frac{r}{a}, 1\right)(13)$$

where $u = (bt)^a$, and through Eq. (10) by replacing $\delta = p/(p-1)$, $c = 1, s = r/a$, the proof can be concluded.

Renyi and Shannon entropies

The entropy of a random variable X may be defined as a measure of uncertainty of a random variable. Entropy is an important concept with many uses in science and engineering. The Renyi entropy[11] is expressed as:

$$I_R(\gamma) = \frac{1}{1-\gamma}\log\left\{\int_{\mathbb{R}} f^\gamma(x)\,dx,\right\}, \quad \gamma > 0 \text{ and } \gamma \neq 1.$$

For the NWG distribution, the following Equation governs:

$$I_R(\gamma) = \frac{1}{1-\gamma}\log\left\{\int_{\mathbb{R}} \frac{(ab^a(1-p))^\gamma\, x^{\gamma(a-1)}e^{-\gamma(bx)^a}}{\left[1-p\left(1-e^{-(bx)^a}\right)\right]^{2\gamma}}dx\right\}$$
$$= \frac{1}{1-\gamma}\log\left\{(ab)^{\gamma-1}(1-p)^\gamma\frac{1}{\Gamma(2\gamma)}\sum_{j=0}^\infty\sum_{k=0}^\infty\frac{(-1)^k p^j\Gamma(2\gamma+j)}{k!(j-k)!}\int_0^\infty y^{(\gamma-1)(1-\frac{1}{a})}e^{-(\gamma+k)y}dy\right\}.$$

However:

$$\int_0^\infty y^{(\gamma-1)\left(1-\frac{1}{a}\right)}e^{-(\gamma+k)y}dy = \frac{\Gamma\left(\gamma-(\gamma-1)\frac{1}{a}\right)}{(\gamma+k)^{\gamma-(\gamma-1)\frac{1}{a}}},$$

And Subsequently:

$$I_R(\gamma) = \frac{1}{1-\gamma}\log\left\{\frac{(ab)^{\gamma-1}(1-p)^{\gamma}\,\Gamma\left(\gamma-(\gamma-1)\frac{1}{a}\right)}{\Gamma(2\gamma)}\sum_{j=0}^{\infty}\sum_{k=0}^{\infty}\frac{(-1)^k p^j\Gamma(2\gamma+j)}{k!\,(j-k)!(\gamma+k)^{\gamma-(\gamma-1)\frac{1}{a}}}\right\}.$$

The Shannon entropy[12] is defined by $E[-\log(f(X))]$. Here we have a particular case derived from $\lim_{\gamma\to1} I_R(\gamma)$. Hence

$$E[-\log(f(X))] = -\log(ab^a(1-p)) - (a-1)E[\log(X)] + E[(bX)^a]$$
$$+2E\left[\log\left\{1-p\left(1-e^{-(bX)^a}\right)\right\}\right].$$

Order statistics

Proposition 3 *It is assumed that* $X_1, X_2, \ldots, X_n$ *are random nonnegative variables and iid such that* $X_i \sim NWG(a,b,p)$. *The density of the* ith *order statistics,* $X_{i:n}$*,is expressed by*

$$f_{i:n}(x) = (1-p)^i\left[1-p\left(1-e^{-(bx)^a}\right)\right]^{-(n+1)}g_{i:n}(x) \tag{12}$$

Where

$$g_{i:n}(x) = \frac{ab^a}{B(i,n-i+1)}x^{a-1}e^{-(n-i+1)(bx)^a}\left[1-e^{-(bx)^a}\right]^{i-1}, \tag{13}$$

is pdf of the ith order statistic of the Weibull (a, b) distribution and $B(n,m) = \int_n^m w^{n-1}(1-w)^{m-1}dw$ is Beta function.

Proof Through defining that $f_{i:n}(y) = n\,f(y)\begin{pmatrix} n-1 \\ i-1 \end{pmatrix}(F(y))^{i-1}(S(y))^{n-1}$. Then,

$$f_{i:n}(x) = \frac{ab^a(1-p)^i}{B(i,n-i+1)}x^{a-1}e^{-(n-i+1)(bx)^a}\,\frac{\left(1-e^{-(bx)^a}\right)^{i-1}}{[1-p(1-e^{-(bx)^a})]^{n+1}}. \tag{14}$$

Equation (14) can be rewritten in terms of $g_{i:n}(x)$ as

$$f_{i:n}(x) = (1-p)^i\left[1-p\left(1-e^{-(bx)^a}\right)\right]^{-(n+1)}g_{i:n}(x). \tag{15}$$

□

Further, the density of $X_{i:n}$ can be written as a combination of the Weibull order statistics densities. Incorporating Eq. (3) in Eq. (12), the following is obtained:

$$f_{i:n}(x) = (1-p)^{i+1}\frac{n!\,(n+j-i)!}{(n-i)!\,(n+j)!}\sum_{j=0}^{\infty}\begin{pmatrix} n+j \\ n \end{pmatrix}\left(\frac{p}{1-p}\right)^j g_{i:n+j}(x). \tag{16}$$

Maximum likelihood estimation

This section discusses the $MLEs$ values of the three unknown parameters of $\theta = (a,b,p)$ of Eq. (3). Supposing $X_1, X_2, \ldots, X_n$ are random samples of size n from $NWG(a,\ b,\ p)$. The log-likelihood function can be expressed in the following form:

$$L = n\log(a) + na\log(b) + n\log(1-p) + (a-1)\sum_{i=1}^{n}\log(x_i) - b^a\sum_{i=1}^{n}x_i^a$$
$$-2\sum_{i=1}^{n}\log\left[1-p\left(1-e^{-(bx_i)^a}\right)\right]. \tag{17}$$

The first-order partial derivatives of Eq. (17) regarding the three unknown parameters are presented by

$$\frac{\partial L}{\partial a} = \frac{n}{a} + n\log(b) + \sum_{i=1}^{n}\log(x_i) - \sum_{i=1}^{n}(bx_i)^a\log(bx_i) + 2p\sum_{i=1}^{n}\frac{(bx_i)^a\log(bx_i)\,e^{-\left(bx_i^a\right)}}{1-p\left(1-e^{-(bx_i)^a}\right)}, \tag{18}$$

$$\frac{\partial L}{\partial b} = \frac{na}{b} - \frac{a}{b}\sum_{i=1}^{n}(bx_i)^a + \frac{2pa}{b}\sum_{i=1}^{n}\frac{(bx_i)^a e^{-(bx_i)^a}}{1-p\left(1-e^{-(bx_i)^a}\right)}, \tag{19}$$

$$\frac{\partial L}{\partial b} = -\frac{n}{1-p} + 2\sum{}_{i=1}^{n} \frac{1-e^{-(bx_i)^a}}{1-p\left(1-e^{-(bx_i)^a}\right)}. \tag{20}$$

By solving this nonlinear system of Eqs. (17)–(19), these solutions will yield the ML estimators for $\widehat{a}, \widehat{b}$ and $\widehat{p}$. For the interval estimate and hypothesis tests of the parameters, it was required that $J(\theta)$ the 3×3 Fisher information matrix be defined by $J(\theta) = E\left[-\frac{\partial^2 L}{\partial\theta_i \partial\theta_j}\right]$. Under regularity conditions, the approximate of the MLEs of θ, the $\widehat{\theta}$, can be approximately calculated as $N_3(0, J^{-1}(\theta))$. The mentioned Equation governs solely for parameters in the parameter space's interior and not on the boundary. The asymptotic distribution of $\sqrt{n}(\widehat{\theta}-\theta)'$ is $N_3(0, J^{-1}(\theta))$, where $J(\theta)$ is the unit information matrix evaluated at $\widehat{\theta}$. This can be utilized to develop the approximate confidence interval for any of the parameters. Therefore, an asymptotic confidence interval with a confidence coefficient $\%100(1-\alpha)$ can be obtained as:

$$ACI_j = (\widehat{\theta}_j \pm z_{1-\frac{\alpha}{2}}\sqrt{\widehat{k}^{\theta_j,\theta_j}}) \tag{21}$$

where $\widehat{k}^{\theta_j,\theta_j}$ denotes the jth diagonal element of $I(\widehat{\theta})^{-1}$ for $j = 1, 2, 3$, and $z_{1-\alpha/2}$ is the $1-\alpha/2$ standard normal quantile.

The EM algorithm

It is assumed that $(X; Z)$ is a random vector, in which X stands for the observed data and Z for the data that is missing. In this stage, a hypothetical complete-data distribution is defined with a joint pdf, through

$$g(x,z;\Theta) = ab^a(1-p)x^{a-1}e^{-(bx)^a}zp^{z-1}(1-e^{-(bx)^a})^{z-1}, \tag{22}$$

where $g(x,z;\Theta)$, $x>0$ and $z \in \mathbb{N}$. The E-step cycle in this formulation will require the expectation of $(Z|X;\Theta^{(r)})$ where $\Theta^{(r)} = (a^{(r)}, b^{(r)}, p^{(r)})$ is the current estimate of Θ in the rth iteration. It must be noted that the pdf of $(Z|X)$, $g(Z|X)$, is given by

$$g(z|x;\Theta) = z\left\{p(1-e^{-(bx)^a})\right\}^{z-1}\left[1-p(1-e^{-(bx)^a})\right]^2, \tag{23}$$

In which $z \in N$. The conditional expectation is therefore expressed by:

$$E[Z|X;\Theta] = \frac{1+p(1-e^{-(bx)^a})}{1-p(1-e^{-(bx)^a})}. \tag{24}$$

The EM cycle will be complete by the M-step process by utilizing the maximum likelihood estimation over Θ. The above-mentioned conditional expectations of the missing values replace them. After disregarding the constants, the log-likelihood function of the complete data can be formulated as:

$$L^* = n\log(ab^a(1-p)) + (1-a)\sum{}_{i=1}^{n}\log(x_i) - \sum{}_{i=1}^{n}(bx_i)^a + \sum{}_{i=1}^{n}(z_i-1)\log\left[p(1-e^{(-bx_i)^a})\right] \tag{25}$$

The first-order partial derivatives of the above Equation concerning the three unknown parameters will be:

$$\frac{\partial L^*}{\partial a} = \frac{n}{a} + n\log(b) + \sum{}_{i=1}^{n}\log(x_i) - \sum{}_{i=1}^{n}(bx_i)^a\log(bx_i)\left[\frac{1-z_ie^{-(bx_i)^a}}{1-e^{-(bx_i)^a}}\right], \tag{26}$$

$$\frac{\partial L^*}{\partial b} = \frac{na}{b} - ab^{a-1}\sum{}_{i=1}^{n}x_i^a\left[\frac{1-z_ie^{-(bx_i)^a}}{1-e^{-(bx_i)^a}}\right], \tag{27}$$

$$\frac{\partial L^*}{\partial p} = -\frac{n}{1-p} + \frac{1}{p}\sum{}_{i=1}^{n}(z_i-1). \tag{28}$$

When the missing Z' values are replaced with their conditional expectations $E[Z|X;\Theta^{(r)}]$, an iterative procedure of the EM algorithm is obtained that is as follows:

$$\widehat{a}^{(r+1)} = -n\left[n\log\left(b^{(r)}\right) + \sum{}_{i=1}^{n}\log(x_i) - \sum{}_{i=1}^{n}\left(b^{(r)}x_i\right)^{a^{(r)}}\log\left(b^{(r)}x_i\right)\left(\frac{1-z_i^{(r)}e^{-\left(b^{(r)}x_i\right)^{a^{(r)}}}}{1-e^{-\left(b^{(r)}x_i\right)^{a^{(r)}}}}\right)\right]^{-1},$$

$$\widehat{b}^{(r+1)} = n\left[\left(b^{(r)}\right)^{a^{(r)}-1}\sum{}_{i=1}^{n}x_i^{a^{(r)}}\left(\frac{1-z_i^{(r)}e^{-\left(b^{(r)}x_i\right)^{a^{(r)}}}}{1-e^{-\left(b^{(r)}x_i\right)^{a^{(r)}}}}\right)\right]^{-1},$$

$$\widehat{p}^{(r+1)} = \frac{\sum{}_{i=1}^{n}(z_i^{(r)}-1)}{n+\sum{}_{i=1}^{n}(z_i^{(r)}-1)},$$

where $\widehat{a}^{(r+1)}$, $\widehat{b}^{(r+1)}$ and $\widehat{p}^{(r+1)}$ are found numerically. Hence, for $i = 1,\ 2, \cdots, n$, in the following equation:

$$z_i^{(r)} = \frac{1 + p^{(r)}(1 - e^{-\left(b^{(r)}x_i\right)^{a^{(r)}}})}{1 - p^{(r)}(1 - e^{-\left(b^{(r)}x_i\right)^{a^{(r)}}})}.$$

Bayesian estimation

This section discusses the problem of Bayesian estimation of the unknown parameters of the NWG distribution. To make the desired inference in the Bayesian estimation, the prior for the unknown parameter and selection of loss function must be considered. Considering the real-world phenomena are flexible and applicable, the following priors for the unknown parameters a, b, and p can be proposed with the following having pdfs:

$$\pi\ (a) \propto\ a^{u_1-1}e^{-v_1 a},\ u_1 > 0,\ v_1 > 0,$$

$$\pi\ (b) \propto\ b^{u_2-1}e^{-v_2 b}, u_2 > 0,\ v_2 > 0,$$

$$\pi\ (p) \propto\ p^{u_3-1}(1-p)^{v_3-1},\ u_3 > 0,\ v_3 > 0.$$

Here, $u_1,\ v_1, u_2,\ v_2, u_3$, and v_3 are the hyper-parameters whose values are accessible from the past available data. As an example, consider M as the number of samples available from NWG(a, b, p) distribution. Now, each sample can have the associated maximum likelihood estimates of (a, b, p) say ($\widehat{a}^i, \widehat{b}^i, \widehat{p}^i$), $i = 1,2, \ldots, M$. Next on equating the mean and variance of these MLEs with the mean and variance of considered gamma priors, the values of hyper-parameters can be obtained. For instance, for the gamma prior of a, Mean(a) is v_1/u_1, whereas Variance(a) is $v_1/{u^2}_1$. The result will be:

$$u_1 = \frac{\left(\frac{1}{M}\sum{}_{i=1}^{M}\widehat{a}^i\right)^2}{\frac{1}{M-1}\sum{}_{i=1}^{M}\left(\widehat{a}^i - \frac{1}{M}\sum{}_{i=1}^{M}\widehat{a}^i\right)^2}, \quad v_1 = \frac{\frac{1}{M}\sum{}_{i=1}^{M}\widehat{a}^i}{\frac{1}{M-1}\sum{}_{i=1}^{M}\left(\widehat{a}^i - \frac{1}{M}\sum{}_{i=1}^{M}\widehat{a}^i\right)^2}.$$

Similarly, the hyper-parameter values of u_2, v_2, u_3 and v_3 can be easily obtained by replacing u_1 with $u_{2,}$, u_3, v_1 with v_2, v_3, and $\widehat{a}^i$ with $\widehat{b}^i$ and $\widehat{p}^i$ in the above expressions. Also, the joint posterior distribution of (a, b, p) has the following form:

$$\pi\ (a, b, p|\boldsymbol{x}) \propto\ a^{u_1+n-1}b^{na+u_2-1}p^{u_3-1}(1-p)^{n+v_3-1}e^{-(v_1 a + v_2 b)}$$

$$\times e^{-\sum{}_{i=1}^{n}(bx_i)^a}\prod{}_{i=1}^{n}\left(\frac{{x_i}^{a-1}}{\left(1 - p(1 - (1 - e^{-(bx_i)^a}))\right)^2}\right)$$

$$\propto\ G_a\left(u_1, v_1\right)G_{b|a}\left(na + u_2, v_2\right)Beta\left(u_3, n + v_3\right)Q\left(a, b\right).$$

Now, different loss functions are discussed. One loss function with the most extensive use is the squared error loss (SEL) function. It is defined as $\delta_{SEL}\left(\psi\ (\varphi), \psi\ (\widehat{\varphi})\right) = \left(\psi\ (\varphi) - \psi\ (\widehat{\varphi})\right)^2$ where $\psi\ (\widehat{\varphi})$ is an estimator of $\psi\ (\varphi)$. Due to its symmetric nature, the SEL function assigns the same weights to overestimation and underestimation. In practice, however, overestimation of a parameter can result in more severe or less severe consequences than underestimation, or vice versa. As a result, an asymmetrical loss function, which favors either overestimation or underestimation, might be a better choice for the estimation of the parameters. A highly practical asymmetric loss function proposed by Varian (1975) is LINEX loss function and is expressed as

$$\delta_{LL}\left(\psi\ (\varphi), \psi\ (\widehat{\varphi})\right) = \exp\left(v\left(\psi\ (\widehat{\varphi}) - \psi\ (\varphi)\right)\right) - h\left(\psi\ (\widehat{\varphi}) - \psi\ (\varphi)\right) - 1; h \neq\ 0. \quad (29)$$

The LINEX loss function is convex and the value of h determines its shape. Note that the magnitude of h in LL shows how asymmetric the function is: if h<0, it shows that the underestimation has a more serious consequence than the overestimation, and the opposite is true for h>0. The Bayes estimator of $\psi\ (\varphi)$ under the LINEX loss function is given by

$$\widehat{\psi}_{LL}(\varphi) = \frac{1}{h}\ln\left(E\left(e^{-h\psi\ (\varphi)}|\boldsymbol{x}\right)\right)$$

Next, samples can be derived from the above posterior density by taking these steps:

Step 1. Generate a_i from $Gamma\left(u_1, v_1\right)$.
Step 2. Having the value of a_i, generate b_i from $Gamma\left(na + u_2, v_2\right)$.
Step 3. Repeat Steps 1 and 2, s times to obtain $\left(a_1, b_1\right), \left(a_2, b_2\right), \ldots, \left(a_s, b_s\right)$.
The Bayes estimate of a under SEL is given by

$$\widehat{\psi}_{SEL}(a) = \frac{\sum{}_{i=1}^{s}\psi\ (a_i)\,Q\left(a_i, b_i\right)}{\sum{}_{i=1}^{s}Q\left(a_i, b_i\right)}$$

and under LINEX by

$$\widehat{\psi}_{LL}(a)=\frac{\sum_{i=1}^{s} e^{-h\psi(a_i)}Q(a_i,b_i)}{\sum_{i=1}^{s} Q(a_i,b_i)}.$$

Now, HPD interval estimate of a can be calculated using the generated importance sample. Consider the posterior density of a, $\pi(a|\boldsymbol{x})$. Suppose that $a^{(k)}=\inf\{a:\pi(a|\boldsymbol{x})\geq k;0<k<1\}$ is the kth quantile of a and $\pi(a|\boldsymbol{x})$ stands for the posterior distribution function of a. For a given a^*, we have $\pi(a^*|\boldsymbol{x})=E[I_{a\leq a^*}(a)|\boldsymbol{x}]$, where $I_{a\leq a^*}(a)$ is the indicator function, written as

$$I_{a\leq a^*}(a)=\begin{cases}1, & a\leq a^*\\ 0, & a>a^*\end{cases}$$

Therefore, a simulation consist estimator of $\pi(a^*|\boldsymbol{x})$ is

$$\widehat{\pi}(a^*|\boldsymbol{x})=\frac{\sum_{i=1}^{s} I_{a\leq a^* Q(a_i,b_i)}}{\sum_{i=1}^{s} Q(a_i,b_i)}.$$

If $a_{(i)}$ be the order values of a_i, define

$$\omega_i=\frac{Q(a_i,b_i)}{\sum_{i=1}^{s} Q(a_i,b_i)},\quad i=1,2,\ldots,s.$$

$\widehat{\pi}(a^*|\boldsymbol{x})$ can be rewritten as

$$\widehat{\pi}(a^*|\boldsymbol{x})=\begin{cases}0, & a^*<a_{(1)}\\ \sum_{j=1}^{i}\omega_j, & a_{(i)}\leq a^*<a_{(i+1)}\\ 1, & a^*>a_{(s)}\end{cases}$$

Therefore, the following will be an approximate estimator of $a^{(k)}$:

$$a^{(k)}=\begin{cases}a_{(1)}, & k=0\\ a_{(i)}, & \sum_{j=1}^{i}\omega_j<k\leq\sum_{j=1}^{i}\omega_j\end{cases}$$

Now, the $100(1-k)\%$, $0<k<1$ confidence intervals of a are formulated as $\left(\widehat{a}^{\left(\frac{j}{M}\right)},\widehat{a}^{\left(\frac{j+[(1-k)s]}{s}\right)}\right)$, $j=1,2,\ldots,s-[(1-k)s]$ with $[u]$ indicate the largest integer less than or equal to u. Ultimately, the HPD credible interval of a is the shortest.

Shrinkage preliminary test estimator

In this section, the shrinkage and preliminary test estimators are acquired. In this regard, it is supposed that pieces of non-sample information in the form of $\theta=\theta_0$ exist and the interest is estimating θ according to the existing information. To determine how accurate the information is, the hypothesis of $H_0:\theta=\theta_0$ and

		$\widehat{a}$	$\widehat{b}$	$\widehat{p}$	
a,b,p	n	Avg MSE	Avg MSE	Avg MSE	
(0.5,1,0.3)	50	0.6631 0.0732	1.6190 0.0420	0.4398	0.1144
	100	0.6359 0.0084	1.6545 0.0398	0.4000	0.1019
	200	0.5982 0.0069	1.6622 0.0390	0.3653	0.0927
	500	0.5544 0.0067	1.4301 0.0371	0.3331	0.0886
(0.5,1,0.4)	50	0.8311 0.0042	1.6080 0.0691	0.4589	0.0225
	100	0.7974 0.0033	1.5888 0.0533	0.4303	0.0148
	200	0.7167 0.0030	1.5103 0.0245	0.4421	0.0140
	500	0.6995 0.0027	1.4992 0.0117	0.4582	0.0011
(2,1,0.5)	50	1.5525 0.0428	1.4163 0.0771	0.5773	0.0280
	100	1.5342 0.0222	1.3950 0.0378	0.5592	0.0139
	200	1.5619 0.0191	1.3968 0.0180	0.5311	0.0115
	500	1.5003 0.0086	1.3833 0.0145	0.5405	0.0063
(2,2,0.6)	50	2.6122 0.0649	2.2222 0.0407	0.6890	0.0383
	100	2.5839 0.0517	1.8945 0.0258	0.6814	0.0335
	200	2.0100 0.0388	1.8881 0.0223	0.6248	0.0204
	500	1.5989 0.0328	1.9819 0.0111	0.6335	0.0197

Table 1. Average (Avg) and MSE of the MLE values for various parameter values.

a,b,p	n	$\widehat{a}$	$\widehat{b}$	$\widehat{p}$
(0.5,1,0,3)	50	(0.1119,0.6418)	(0.9054,1.3828)	(0.1237,0.3579)
	100	(0.2599,0.6170)	(0.8735,1.4777)	(0.1619,0.3352)
	200	(0.2695,0.5348)	(0.8711,1.3152)	(0.1878,0.3196)
	500	(0.3344,0.6741)	(0.8886,1.3102)	(0.2001,0.3239)
(0.5,1,0,4)	50	(0.4438,0.6922)	(0.8632,1.2817)	(0.2423,0.4783)
	100	(0.3647,0.6639)	(0.8195,1.3303)	(0.2222,0.4685)
	200	(0.2650,0.6652)	(0.9011,1.4288)	(0.3101,0.4501)
	500	(0.3766,0.6722)	(0.9144,1.4309)	(0.3200,0.4414)
(2,1,0,5)	50	(1.6273,2.2770)	(0.7745,1.5871)	(0.3427,0.6698)
	100	(1.6845,2.5011)	(0.7912,1.3902)	(0.3693,0.6239)
	200	(1.1361,2.1317)	(0.8326,1.3052)	(0.4928,0.6071)
	500	(1.7294,2.1030)	(0.8120,1.1858)	(0.4771,0.6002)
(2,2,0,6)	50	(1.5591,2.4930)	(1.8601,2.5744)	(0.5242,0.8088)
	100	(1.5848,2.3152)	(1.8890,2.4343)	(0.5511,0.8799)
	200	(1.6423,2.2845)	(1.7031,2.3310)	(0.5553,0.7348)
	500	(1.6061,2.1561)	(1.7222,2.3011)	(0.5961,0.7199)

Table 2. %95 interval estimates for various parameter values.

		$\widehat{a}^{SPT}$	$\widehat{b}^{SPT}$	$\widehat{p}^{SPT}$
a,b,p	n	**Avg MSE**	**Avg MSE**	**Avg MSE**
(0.5,1,0.3)	50	0.7091 0.0654	1.2553 0.0302	0.2476 0.0991
	100	0.6714 0.0079	1.2346 0.0021	0.2897 0.0073
	200	0.6608 0.0066	1.1178 0.0015	0.3271 0.0058
	500	0.6332 0.0052	1.2014 0.0001	0.3386 0.0051
(0.5,1,0.4)	50	0.5885 0.0037	1.1644 0.0047	0.5872 0.0231
	100	0.5911 0.0029	1.2718 0.0041	0.5669 0.0136
	200	0.6312 0.0020	1.2007 0.0038	0.5111 0.0100
	500	0.6097 0.0016	1.2509 0.0030	0.4827 0.0001
(2,1,0.5)	50	1.8928 0.0382	1.5032 0.0093	0.6090 0.0158
	100	1.9099 0.0200	1.4980 0.0081	0.6221 0.0041
	200	1.9522 0.0159	1.4990 0.0075	0.5583 0.0038
	500	1.9134 0.0114	1.2483 0.0071	0.6065 0.0034
(2,2,0.6)	50	1.7672 0.0555	2.0010 0.0023	0.6722 0.0030
	100	1.4777 0.0065	2.3188 0.0021	0.6432 0.0022
	200	1.6680 0.0041	2.1922 0.0019	0.6515 0.0016
	500	1.7993 0.0025	2.1090 0.0016	0.6793 0.0013

Table 3. Average (Avg) and MSE values of the MLE SPT for various parameter values.

alternate $H_1 : \theta \neq \theta_0$ may be considered; however, it has been observed that the construction of shrinkage estimators for θ based on fixed alternatives $H_1 : \theta = \theta_0 + \xi$ for a fixed ξ does not have a significant effect on the performance $H_1 : \theta = \theta_0 + \xi$ in comparison with $\widehat{\theta}$. It also means that the asymptotic distribution of shrinkage estimator is in correspondence with that of $\widehat{\theta}$ in Saleh study (Saleh, 2013). Therefore, to overcome such challenge, the present study suggests local alternatives with the form of $A_{(n)} : \theta_n = \theta_0 + n^{-1/2}\xi$, where ξ is a fixed number. Thus, the Shrinkage Preliminary Test (SPT) estimator of θ based on maximum likelihood estimators can be expressed as

$$\widehat{\theta}_{MLE.SPT} = \mu\,\theta_0 + (1-\mu)\,\widehat{\theta}_{MLE} I\left(\Lambda_n < \chi^2_1(\gamma)\right),$$

and

$$\widehat{\theta}_{Bayes.SPT} = \mu\,\theta_0 + (1-\mu)\,\widehat{\theta}_{Bayes} I\left(\Lambda_n < \chi^2_1(\gamma)\right).$$

			$\widehat{a}$	$\widehat{b}$	$\widehat{p}$	
a, b, p		n	Avg MSE	Avg MSE	Avg MSE	
(0.5,1,0.3)	SEL	50	0.5722 0.0534	1.2157 0.0305	0.3839 0.0387	0.0387
		100	0.5501 0.0077	1.3390 0.0288	0.3912 0.0303	0.0303
		200	0.5111 0.0060	1.2886 0.0255	0.3774	0.0205
		500	0.5278 0.0063	1.1626 0.0241	0.3355	0.0122
	LINEX	50	0.5314 0.0087	1.0720 0.0063	0.3666	0.0096
		100	0.5002 0.0053	1.0053 0.0060	0.3725	0.0092
		200	0.4716 0.0047	1.0066 0.0039	0.3511	0.0085
		500	0.5024 0.0031	1.0063 0.0035	0.3010	0.0079
(0.5,1,0.4)	SEL	50	0.8543 0.0040	1.0521 0.0088	0.4136	0.0188
		100	0.7395 0.0028	1.0373 0.0072	0.4020	0.0074
		200	0.7713 0.0027	1.0400 0.0069	0.4273	0.0071
		500	0.6851 0.0025	1.0311 0.0063	0.4004	0.0061
	LINEX	50	0.8883 0.0007	1.0023 0.0005	0.4002	0.0043
		100	0.9709 0.0006	1.0019 0.0005	0.3977	0.0038
		200	0.9538 0.0005	1.0016 0.0003	0.3990	0.0037
		500	0.8425 0.0005	1.0005 0.0002	0.4156	0.0029
(2,1,0.5)	SEL	50	1.6812 0.0300	0.8890 0.0065	0.5155	0.0153
		100	1.3191 0.0210	0.8992 0.0058	0.5062	0.0019
		200	1.5208 0.0133	0.9003 0.0057	0.5211	0.0014
		500	1.5577 0.0057	0.9517 0.0054	0.5130	0.0013
	LINEX	50	1.9222 0.0077	0.7325 0.0041	0.5050	0.0011
		100	1.9738 0.0069	0.7610 0.0033	0.4998	0.0009
		200	1.9545 0.0066	0.8020 0.0030	0.5041	0.0083
		500	1.9497 0.0065	0.8223 0.0028	0.5001	0.0078
(2,2,0.6)	SEL	50	2.0014 0.0472	2.1121 0.0302	0.6060	0.0008
		100	2.2373 0.0051	2.0015 0.0127	0.6181	0.0005
		200	2.9948 0.0049	1.9888 0.0111	0.6053	0.0005
		500	2.0160 0.0022	1.9890 0.0092	0.6002	0.0005
	LINEX	50	1.8854 0.0091	2.0531 0.0245	0.6002	0.0004
		100	1.9002 0.0008	2.0002 0.0103	0.6113	0.0006
		200	2.0005 0.0006	1.9993 0.0084	0.6014	0.0003
		500	1.9723 0.0006	2.0011 0.0080	0.6002	0.0003

Table 4. Average (Avg) and MSE of the Bayes values for various parameter values.

In the mentioned equation, $\mu \in [0, 1]$. Moreover, it is stated that under the condition of H_0, $\sqrt{n(\theta - \theta_0)}$ is asymptotically $N(0, var\left(\widehat{\theta}\right))$, consequently, the test statistics are represented as $\Lambda_n = \left(\sqrt{n}\left(\widehat{\theta} - \theta\right)\right)^2 / var\left(\widehat{\theta}\right)$. Afterward, the asymptotic distribution of Λ_n converges to a non-central chi-square distribution having one degree of freedom and noncentrality parameter of $\Delta^2/2$, where $\Delta^2 = \xi^2/\xi^2\left(\widehat{\theta}\right)$. As a result, the H_0 is rejected, when $\Lambda_n > \chi_1^2(\alpha)$ where α is type-I error and $\chi_1^2(\alpha)$ is the α-upper quantile of Chi-square distribution having one degree of freedom. The SPT estimators for all parameters based on maximum likelihood and a similar procedure can be used to define Bayesian estimators.

Simulation study

Here, the maximum likelihood and Bayesian estimation method's performance to estimate the parameters of the NWG are investigated using Monte-Carlo simulation for an *n* number of different sample sizes and different parameter values. The simulation was performed using R software. It involves 1000 replications for different sample sizes of 50,100,200,500, and different parameter combinations. The following summarizes the Monte-Carlo algorithm:

1- Generate the values $x_1, x_2, \ldots, x_n$ from the $NWG(\alpha, \beta, p)$ using the Eq. (3).

2- Estimate the parameters of the NWG by the MLE method.

3-Perform 1000 replications of steps (1) and (2).

4- Compute the mean and mean square error of the 1000 parameter estimates for each of the three parameters of the NWG. The analytical expressions of these statistics are obtained by $\overline{\widehat{\theta}} = \frac{1}{1000}\sum_{i=1}^{1000}\widehat{\theta}_i$,

			$\widehat{a}$	$\widehat{b}$	$\widehat{p}$	
a, b, p		n	Avg MSE	Avg MSE	Avg MSE	
(0.5,1,0.3)	SEL	50	0.5311 0.0025	1.1835 0.0188	0.3541 0.0387	0.0275
		100	0.5491 0.0020	1.2310 0.0090	0.3302 0.0303	0.0212
		200	0.5018 0.0016	1.2114 0.0084	0.3214	0.0134
		500	0.5128 0.0014	1.1466 0.0081	0.3015	0.0101
	LINEX	50	0.5302 0.0067	1.0020 0.0093	0.3473	0.0088
		100	0.5001 0.0033	1.0012 0.0054	0.3222	0.0082
		200	0.4877 0.0026	1.0009 0.0047	0.3101	0.0075
		500	0.4936 0.0020	1.0007 0.0055	0.3003	0.0066
(0.5,1,0.4)	SEL	50	0.8090 0.0037	1.0377 0.0064	0.4021	0.0134
		100	0.6821 0.0036	1.0138 0.0062	0.4018	0.0064
		200	0.6980 0.0029	1.0202 0.0057	0.4166	0.0062
		500	0.6653 0.0023	1.0131 0.0055	0.4003	0.0058
	LINEX	50	0.8122 0.0005	1.0024 0.0004	0.4004	0.0041
		100	0.9322 0.0005	1.0015 0.0004	0.3979	0.0035
		200	0.9983 0.0004	1.0010 0.0003	0.3885	0.0031
		500	0.9415 0.0002	1.0004 0.0001	0.4075	0.0028
(2,1,0.5)	SEL	50	1.1831 0.0214	0.9780 0.0058	0.5035	0.0023
		100	1.2133 0.0202	0.8337 0.0053	0.5021	0.0018
		200	1.3276 0.0119	0.9133 0.0052	0.5113	0.0016
		500	1.0537 0.0053	0.9575 0.0046	0.5122	0.0010
	LINEX	50	1.2502 0.0071	0.8381 0.0043	0.5037	0.0009
		100	1.1234 0.0044	0.8621 0.0038	0.4884	0.0008
		200	1.1555 0.0041	0.8998 0.0029	0.4905	0.0062
		500	1.1417 0.0038	0.9213 0.0024	0.5021	0.0058
(2,2,0.6)	SEL	50	2.0008 0.0226	2.1002 0.0202	0.6113	0.0007
		100	2.1150 0.0045	2.0022 0.0102	0.6024	0.0005
		200	2.2642 0.0040	1.9969 0.0098	0.6016	0.0004
		500	2.0040 0.0036	1.9880 0.0090	0.6002	0.0004
	LINEX	50	1.9866 0.0081	2.0031 0.0153	0.6001	0.0003
		100	1.9782 0.0007	2.0005 0.0093 00.010\\0.0098	0.6003	0.0005
		200	2.0001 0.0006	1.9885 0.0081	0.6004	0.0002
		500	1.9696 0.0005	2.0005 0.0077	0.6002	0.0002

Table 5. Average (Avg) and MSE values of the bayesian SPT for various parameter values.

and $MSE(\theta_i) = \frac{1}{1000}\sum_{j=1}^{1000}(\widehat{\theta}_{ij} - \theta_i),^2$ where $\widehat{\theta}_{ij}$ denotes the MLE of θ_i in the jth replication, for $i = 1, 2, 3$ and $j = 1, 2, \cdots, 1000$.

The MLE values, %95 confidence interval estimation and the mean squared error (MSE) of the parameter estimates are listed in Tables 1 and 2. Notice that the MLE method performs acceptability for estimating the model parameters. Subsequently, the test statistic of Λ_n is first under the null hypothesis, $H_0 : \theta = \theta_0$ to find shrinkage preliminary test estimates. It is considered that the $\mu = 0.7$ and the type-I error is 0.05. Afterward, maximum likelihood estimators given by the EM algorithm are exploited for maximum likelihood based shrinkage preliminary test. Table 3 shows all the estimates. Also, it can be seen that the MSE of the shrinkage preliminary test estimators are lower than the maximum likelihood-based estimators. Additionally, the Bayesian and shrinkage estimators are tabulated in Tables 4 and 5. It is observed that Bayes estimators have more or less MSE than MLE estimators. %95 HPD confidence interval estimations are also computed in Table 6. It is observed that HPD intervals have smaller length than asymptotic intervals.

Tables 1 and 2 present the results of the Monte Carlo simulation for different parameter settings (α, β, p) and sample sizes n = 50, 100, 200, 500. Table 1 shows the averages of the maximum likelihood estimates (MLEs) and their mean squared errors (MSEs) for each parameter. As expected, as the sample size increases, the estimates get closer to the true values, and the MSEs decrease, indicating improved estimation accuracy.

For example, for the parameter combination (0.5, 1, 0.3), the estimates approach the true parameter values as n grows. This trend is consistent for other parameter sets as well, confirming the good performance of the MLE method.

Table 2 reports the 95% confidence intervals of the MLEs. The interval lengths decrease as the sample size increases, improving estimation precision. Most intervals cover the true parameters, demonstrating good coverage properties. For instance, in the case (2, 2, 0.6), the interval lengths clearly shrink from n = 50 to n = 500.

a, b, p	n	$\widehat{a}$	$\widehat{b}$	$\widehat{p}$
(0.5,1,0.3)	50	(0.0885,0.5473)	(0.9334,1.2816)	(0.1307,0.2998)
	100	(0.2337,0.5883)	(0.9025,1.4777)	(0.1929,0.3052)
	200	(0.2351,0.5034)	(0.8711,1.2242)	(0.1878,0.3196)
	500	(0.3614,0.5921)	(0.9186,1.1202)	(0.2001,0.3239)
(0.5,1,0.4)	50	(0.4392,0.6612)	(0.9125,1.1927)	(0.2423,0.4783)
	100	(0.3327,0.5726)	(0.9295,1.2013)	(0.3414,0.4188)
	200	(0.2853,0.5955)	(0.9011,1.4288)	(0.3701,0.4404)
	500	(0.3804,0.6323)	(0.9144,1.4309)	(0.3801,0.4556)
(2,1,0.5)	50	(1.7222,2.2393)	(0.8745,1.4297)	(0.3522,0.6435)
	100	(1.6733,2.4428)	(0.8713,1.2312)	(0.4625,0.7291)
	200	(1.0921,2.1002)	(0.9124,1.1031)	(0.4823,0.6051)
	500	(1.8312,2.1140)	(0.9000,1.07323)	(0.4913,0.5822)
(2,2,0,6)	50	(1.5992,2.3927)	(1.9720,2.5744)	(0.5324,0.7888)
	100	(1.5944,2.4172)	(1.9890,2.5225)	(0.5601,0.8699)
	200	(1.7423,2.3894)	(1.8868,2.1322)	(0.5855,0.7245)
	500	(1.6902,2.2592)	(1.9334,2.2419)	(0.6762,0.7107)

Table 6. %95 HPD interval estimates for various parameter values.

Model	NLC	AIC	BIC	KS
NWG	−410.0921	826.1842	834.7403	0.0256
WG	−411.1887	828.1098	835.3943	0.0409
Weibull	−414.0869	832.1738	837.8778	0.0700
GE	−413.0776	830.1552	835.8593	0.0725

Table 7. Goodness-of-fit tests for the real data set.

Overall, these results indicate that the proposed MLE method provides reliable and accurate parameter estimates for the NWG distribution, especially with larger sample sizes.

Application

In this section, the performance of the NWG distribution to a real data on remission time is illustrated in months of random samples of 128 patients with bladder cancer. The real dataset used in this section corresponds to remission times (in months) of 128 patients diagnosed with bladder cancer. This dataset was originally reported by Lee et al.[13] and has been widely used in survival analysis and reliability modeling literature to evaluate and compare statistical distributions. Each observation represents the time (in months) until cancer remission for a patient following treatment. The data includes both short-term and long-term remission times, with values ranging from as low as 0.08 months to as high as 79.05 months.

The purpose of using this real dataset in our study is to assess the practical applicability and flexibility of the proposed NWG distribution in modeling real-world survival data. By fitting the NWG model and comparing it with other established distributions (e.g., Weibull, Weibull-Geometric, and Generalized Exponential), we aim to demonstrate the superiority of the NWG model in terms of goodness-of-fit measures such as AIC, BIC, and KS statistics, as well as through graphical analyses. The data have been reported by Lee et al.[13] and presented below:

0.08, 2.09, 3.48, 4.87, 6.94, 8.66, 13.11, 23.63, 0.20, 2.23, 3.52, 4.98, 6.97, 9.02, 13.29, 0.40, 2.26, 3.57, 5.06, 7.09, 9.22, 13.80, 25.74, 0.50, 2.46, 3.64,5.09, 7.26, 9.47, 14.24, 25.82, 0.51, 2.54, 3.70, 5.17, 7.28, 9.74, 14.76, 26.31, 0.81, 2.62,3.82, 5.32, 7.32, 10.06, 14.77, 32.15, 2.64, 3.88, 5.32, 7.39, 10.34, 14.83, 34.26, 0.90, 2.69, 4.18, 5.34, 7.59, 10.66, 15.96, 36.66, 1.05, 2.69, 4.23, 5.41, 7.62, 10.75, 16.62, 43.01, 1.19, 2.75, 4.26, 5.41, 7.63, 17.12, 46.12, 1.26, 2.83, 4.33, 5.49, 7.66, 11.25, 17.14, 79.05, 1.35, 2.87, 5.62, 7.87, 11.64, 17.36, 1.40, 3.02, 4.34, 5.71, 7.93, 11.79, 18.10, 1.46, 4.40, 5.85, 8.26, 11.98, 19.13, 1.76, 3.25, 4.50, 6.25, 8.37, 12.02, 2.02, 3.31, 4.51, 6.54, 8.53, 12.03, 20.28, 2.02, 3.36, 6.76, 12.07, 21.73, 2.07, 3.36, 6.93, 8.65, 12.63, 22.69.

In this regard, the NWG is compared with the Weibull, Weibull-Geometric and generalized exponential (GE) distribution. The maximum likelihood estimates, Kolmogorov-Smirnov test statistic (KS), Akaike Information Criterion (AIC), and Bayesian Information Criterion (BIC) for the fitted distributions have been displayed in Table 7. It is observed that the NWG distribution results in the smallest value for each of the criteria, implying that the NWG distribution performs better compared to the other models. The MLE values and SPT estimators of a, b, p are obtained via maximizing the log-likelihood function in Eqs. (3) and (31) with respect to three parameters that they are as: $\widehat{a} = 3.7032$, $\widehat{b} = 1.9846$, $\widehat{p} = 0.7001$ and $\widehat{a}^{SPT} = 3.0012$, $\widehat{b}^{SPT} = 0.7938$, $\widehat{p}^{SPT} = 0.5633$. Also, the Bayesian estimators of these three parameters are obtained as $\widehat{a} = 2.9614$, $\widehat{b} = 1.3553$, $\widehat{p} = 0.5104$ and $\widehat{a}^{SPT} = 2.0653$, $\widehat{b}^{SPT} = 0.9825$, $\widehat{p}^{SPT} = 0.3517$.

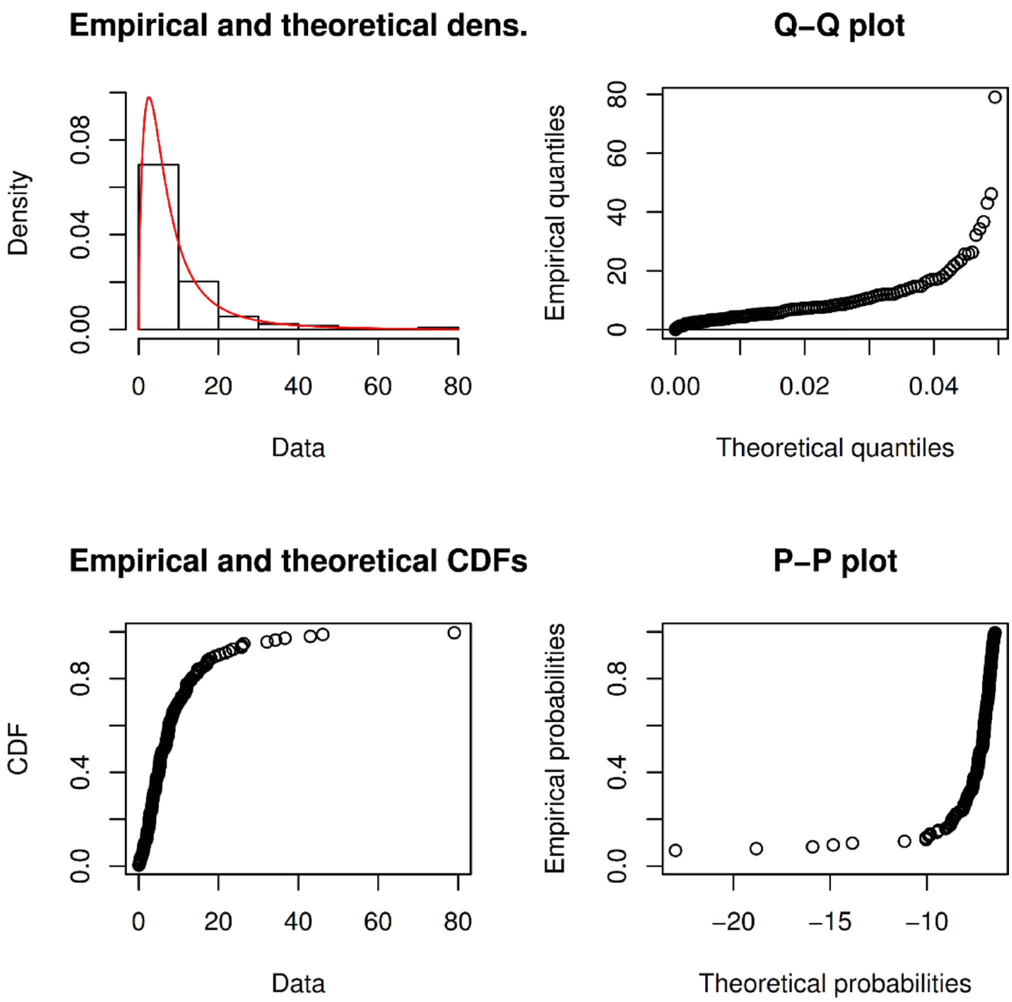

Fig. 3. Fitted density and distribution functions of NWG distribution for the data set.

The estimated pdf of NWG is compared with the histogram of the data set in Fig. 3. It appears that it is reasonable to use the NWG model to analyze the data set.

Figure 3 includes four diagnostic plots that together evaluate the goodness-of-fit of the NWG distribution to the real data on remission times of bladder cancer patients.

Top-left plot shows the fitted probability density function (PDF) of the NWG model against the empirical histogram of the data. The close alignment suggests that the NWG distribution effectively captures the shape and skewness of the data.

Top-right plot presents the Q-Q plot (quantile-quantile plot) comparing the empirical and theoretical quantiles of the fitted NWG distribution. The upward-sloping curve with increasing steepness indicates a heavier right tail in the data, and overall agreement between theoretical and empirical quantiles.

Bottom-left plot is the P-P plot (probability-probability plot), where the theoretical CDF values are plotted against the observed data. The exponential-like curve indicates that the NWG model aligns with the cumulative structure of the dataset, especially in the early stage of remission.

Bottom-right plot compares theoretical probabilities to empirical ones. The gradually increasing curve supports the NWG model's adequacy in capturing the probability structure, especially toward the higher range of remission times.

These plots jointly confirm that the NWG distribution fits the real dataset well in both central and tail regions.

Conclusions

This paper introduces a NWG distribution through compounding the Weibull distribution and geometric distribution. The failure rate functions of this new compound distribution have various shapes; it can be increasing, decreasing, or unimodal. Also, the rth moments and quantiles of the suggested distribution are presented. Additionally, the maximum likelihood estimation method incorporating the EM algorithm is constructed to estimate the NWG distribution parameters. The Fisher information matrix and asymptotic confidence intervals for the MLEs are also provided. Moreover, the Markov Chain Monte Carlo (MCMC) method is applied to obtain point estimators of the unknown parameters under different symmetric and asymmetric loss functions. The highest posterior density (HPD) credible intervals are obtained using the importance sampling procedure. Shrinkage preliminary test estimators are also computed based on maximum likelihood and Bayesian estimators.

Furthermore, a real data set is analyzed for illustration. The results indicate that the NWG distribution provides a better fit compared to several common lifetime distributions. Therefore, the NWG distribution can effectively be used to model data in fields such as reliability, biological systems, and engineering[14,15].

Data availability

All data generated or analyzed during this study are included in this published article.

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Author contributions

A. Zanboori developed the core methodology, derived the theoretical results, and prepared the initial draft of the manuscript. E. Zanboori performed the simulation studies, contributed to data visualization, and managed the revision and submission process. M. Mahmoudi conducted the real data analysis, reviewed the manuscript thoroughly, and provided critical feedback for improvement. H. Parvin reviewed the manuscript thoroughly, and provided critical feedback for improvement. All authors read and approved the final version of the manuscript.

Declarations

Competing Interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to E.Z. or M.M.